The Open Source Firefighter

A coordinated ember defense for the places we live

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> **A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.** *Document ID:* ***OSF-3*** *· one document of the Open Source Firefighter specification family.*

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This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back. Former: City of Delta Junction VFD Assistant Chief · Donnelly Training Area (DMPBAX) Range Manager · U.S. Air Force Tech Control

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The problem is embers

When a wildfire destroys a neighborhood, it’s usually not the wall of flame that does it. It’s embers — burning fragments carried by the wind, sometimes a mile or more ahead of the fire itself. They land on roofs, in gutters, in the dry mulch beside a wall, and they start small fires that become house fires while the main fire is still far away. In a bad wind event, thousands of embers land across a neighborhood at once, and there are never enough firefighters to stand at every house.

There is a well-understood answer: keep the surfaces wet. A roof that stays damp during the ember window doesn’t ignite. Sprinkler systems that wet roofs and perimeters during fire events have been protecting individual homes and cabins for years. The engineering isn’t exotic — it’s water, pipes, and timing.

What’s missing is coordination. If every house on a street turns its sprinklers on at full blast at the same time, the water system that supplies all of them collapses — pressure drops, and nobody’s sprinklers work. A neighborhood that wants to defend itself has to share its water intelligently: houses taking turns, surfaces staying wet, the supply never exhausted. No product on the market does this. That’s the gap the Open Source Firefighter fills.

The box

The heart of the system is a weatherproof box about the size of a breaker panel, mounted on a wall. Inside is an industrial controller, a small mesh radio, a battery, and three motorized water valves. A garden hose or pipe comes in one side; up to three go out the other, feeding ordinary lawn sprinklers placed on the roof line and around the house.

It works like this:

It takes turns. The box opens one water zone at a time, rotating on a schedule, so an ordinary household water connection — the same pressure that runs your garden hose — can keep a whole house wet. A wet roof stays protective for many minutes between passes; embers need dry surfaces, and they don’t get any. This also means the system works with as little as one sprinkler from the hardware store. Two or three can protect a mobile home, a cabin, or a small house.

It talks to the neighbors. Each box carries a small license-free mesh radio. When several homes on a street have them, the boxes coordinate automatically — scheduling their water use so the shared supply is never overwhelmed and no house goes unprotected. The same taking-turns logic that runs inside one box runs across the whole neighborhood. No internet connection is required. No subscription. No account. The boxes talk directly to each other.

It keeps its own records. Every box keeps a tamper-evident log of what it did and when — when it activated, how much water it used, what the conditions were. After a fire, the neighborhood has a verifiable record of what happened, useful for insurance, for fire investigators, and for making the system better.

It works alone. If a box is the only one on the street, or the radio network is down, it simply protects its own house on its own schedule. Coordination makes it better; nothing makes it stop.

It talks to you. Every unit speaks its status aloud in plain language — “system armed,” “zone two fault,” “low battery” — the way a fire alarm panel does, so nobody needs a manual or an app to know what’s happening. The rugged field version goes further: it talks a first-time user through its own setup, step by step, confirming each connection as it’s made. A volunteer who has never seen one can put it into service in minutes, in the dark, in gloves.

The firefighter is in charge. When professional fire crews arrive, the system can hand them a single unified picture of every participating home — what’s running, what water is available — and accept their direction. And a homeowner always retains control of their own equipment. The system’s rule, written into its founding documents: automation manages resources; it never overrules people.

What it deliberately is not

The Open Source Firefighter is not a smart home system, not a security camera, and not a surveillance network. Its radio carries only short, defined messages about fire defense — activations, water status, coordination. Video, audio, and personal data have no place in its language at all; the system is physically and architecturally incapable of transmitting them. A homeowner’s data stays home. This isn’t a policy promise that could be revised later — it’s built into how the system works, and the program’s charter forbids changing it.

The family

One system, three packages:

Residential. The wall-mounted box described above. Built entirely from standard industrial parts — the same controllers, valves, and batteries used in water treatment plants and fire alarm panels, available from ordinary suppliers. Parts cost is roughly $800–1,500, and a capable homeowner, electrician, or irrigation contractor can build and install it. Every wear item is a commodity: when the battery dies in eight years, the replacement is an $18 brick from any alarm supply house.

Government / Commercial. The same unit in tougher packaging — sealed military-style connectors, marine-grade fittings, higher-end radios where agencies need them, and integration with the mapping and command tools firefighters already use. A participating facility appears on a fire crew’s tablet as a known asset: water, power, communications, and status, at a fixed point.

Tactical. The same system in a rugged transit case with battery and solar power, designed to be stored on a pallet in an emergency warehouse and dropped where it’s needed. Forestry crews already protect threatened structures by setting up portable pumps and sprinklers at remote sites and leaving them running. The tactical unit ties into that existing practice without modifying any of it: it watches the pump by listening to it, monitors the hoses, schedules the water so the fuel lasts the whole event instead of running dry, and reports everything back in real time — so a crew doesn’t drive two hours to discover the pump quit six hours ago. Add a weather station and a thermal detector, and the kit becomes something the fire service lost decades ago and never replaced: a fire lookout — except this one is automated, and it can defend its own ground. Every tactical unit also carries an emergency button that signals for help by four independent paths, including two satellite networks, and an interior fast-charging provision, because a firefighter with a dead phone at night should be able to charge it from the rig. That is an operational capability, not an amenity.

The seatbelt principle

This system will be asked to prove itself before it’s allowed to exist — studies, certifications, market validation, professional sign-off. That demand deserves a direct answer, because history has already graded it.

The Open Source Firefighter is not a product competing for a budget. It is a safety primitive, like the seatbelt or the smoke detector: something whose worth is categorical, whose mechanism is legible from first principles, and whose delay is paid in lives. Both the seatbelt and the smoke detector were made to run the proof gauntlet before adoption, and the studies that eventually arrived mostly measured the cost of the waiting. The honest path for a safety primitive is the one the smoke detector finally took: reason the mechanism, build it, put it in front of the real thing, let it fail occasionally, and improve it in contact with reality — rigorously, in the open, while building.

Two things make that path even safer here than it was for the seatbelt. First, the baseline is zero: a unit that faults or runs dry degrades to exactly what the house had before — nothing. The system is strictly additive by architecture, so occasional failure returns to the unprotected status quo everyone currently accepts without a study. Demanding proof before a wet roof is allowed to exist is defending dry roofs as the evidentiary default. Second, the proof machine ships inside every box: the verifiable event log means every deployment is a public experiment — what ran, when, with what water, to what outcome — checkable by strangers. The answer to “where’s the study?” is: the study instrument is built in. Help us run it.

One more thing has changed since the seatbelt's day, and it is why this project is structured the way it is. Safety primitives used to get a grace period — time to become culture before anyone figured out how to meter them. The smoke detector arrived before the subscription business model that would have eaten it. That grace period is gone: invented today, the smoke detector would arrive as a platform, with a monthly fee, a proprietary battery, a cloud dependency, and an insurance partnership — and the enclosure would masquerade as the safety. The coordination primitive for ember defense is being born right now, and whichever form entrenches first is what "ember defense" means for the next fifty years. This project's openness is not idealism. It is a replacement for the historical luck that seatbelts got for free.

The goal was never to market the seatbelt. It is to establish the floor — after which building in fire country without one becomes unthinkable.

The program

The Open Source Firefighter is not a product for sale. It is a **published open specification** — a complete, freely available description of how to build compliant equipment, which anyone may use: homeowners, contractors, fire departments, manufacturers, and government agencies, without license fees or permission.

This is deliberate, and it comes from watching what usually happens to good ideas in emergency services equipment. An inventor builds a device; the company fails or is acquired; years later the ideas reappear inside proprietary equipment that can't be repaired without a service contract, with batteries that cost ten times what they should because only one company makes them. The knowledge survives. The repairability dies. The people who depend on the equipment pay for that forever.

The Open Source Firefighter is structured so that can't happen to it:

- **The specification is the invention, and it's given away.** The intellectual property strategy exists to keep the coordination system permanently open — so that no one can ever fence it off and sell it back as a subscription.
- **Everything is a standard part.** The specification requires that every component be available from multiple manufacturers through normal supply channels. Equipment that depends on a single company for parts is, in the specification's own words, expressly non-compliant.
- **It is governed like the internet's own standards.** Anyone may propose improvements. Proposals are judged in public against a short charter whose two unbreakable rules are: the system must never require an outside service to do its job, and automation must never overrule the person standing at the equipment. New technology is welcomed — as optional additions, never as new requirements. A box built to this specification in 2026 and one built in 2046 will still work together.

What one street can do

A single box protects a single home — even a single-wide on a hose bib with one dollar-store sprinkler. A street of them protects each other, sharing water no single home could manage alone. A valley of them, with one satellite link on a ridge, becomes visible to incident commanders as an organized, self-defending community with verifiable records. And every home that defends itself is a home a fire crew doesn't have to stand in front of — which is the real point. The system exists to subtract from the list of places we send people into the ember storm.

At ordinary suburban spacing — houses fifteen to twenty feet apart, a wooden fence bridging the gap — fire science has a blunt finding: once one house lights, the next house is the fuel. Your survival is partly a function of parcels you don't control. That's why participation matters more than perfection: three protected homes out of five is a firebreak; one is a wet spot. And coverage doesn't have to stop at the property line — two hundred feet of good garden hose and a couple of well-chosen small sprinklers will keep the vacant lot between neighbors wet. Not well — enough. The gap in the neighborhood's defense is the neighborhood's problem, and now the neighborhood has a tool shaped like the gap.

This county already ran the experiment. The Black Forest fire of 2013 — a few miles from where this specification was written — destroyed nearly five hundred homes and killed two people. The homes that had done mitigation survived at higher rates, and it still wasn't enough, because homes ignited each other faster than individual preparation could answer. The research since has reached a conclusion the fire service now states plainly: the community, not the house, is the unit of survival. What never existed was infrastructure for a community to act as one. That is the entire reason this system exists.

The wetting schedule itself speaks the fire service's language: zones are classed by the federal fuel-moisture system firefighters already use — grass is a one-hour fuel; a fire-rated roof's real fuel is the debris in its gutters — and the system re-wets each zone on the actual drying rate of its fuel under live weather, from a local sensor, a neighbor's, or the public fire-weather feeds. Smart water, spent exactly as fast as the physics demands and no faster.

What one crew can do

Initial attack has always required putting people where the danger is, because eyes and water both required bodies at the point of contact. This system separates them. Picture a neighborhood with a few dozen boxes, a small drone overhead watching for spot fires, and a crew coordinating from a truck at a safe anchor point — a package small enough to be one engine, one trailer of cases, and a handful of people, deployable like any other fire resource: the boxes are the hands, the drone is the eyes, and the people direct water across the whole neighborhood without standing in the ember cast. A handful of firefighters becomes the initial attack for an entire community — from cover.

The same idea scales past the neighborhood. Where a fire is coming days out — and a century of fire history tells us, for most ground, exactly where it will run — nodes can be placed ahead of it on the land itself: the drainages, the field edges, the chokepoints a fire has to cross. A pre-wetted strip of ground is a fire line that builds itself and needs no crew to hold it, and unlike a bulldozed line it does the job by making the fuel briefly too wet to burn instead of by destroying it. When the fire has passed, the water dries, the vegetation lives, and the town is still a town — not a place that was torn up to be saved. It works against grass and surface fire in the flanking and backing phases where most ground is actually held; it is not a wall against a wind-driven crown run, and it makes no such claim. It is held and commanded by people, always — seasonal crews and their leaders, doing lighter and safer work than cutting hand line, because the water holds the line they would otherwise dig.

That is the heart of it: this is not machines replacing firefighters. It is a technical advantage handed to the firefighting community that already exists — reach and leverage for the people who do the work, built by one of them. When it’s over, the crew goes home and leaves nothing behind but wet ground drying in the sun.

No system prevents every tragedy, and this one makes no such claim. Wildland firefighting will remain dangerous work done by people of uncommon courage. What this system changes is what we have to ask of them — how many of the places on fire require a person standing there. The American West is getting drier. There will be more fire, and there will never be enough crews. The honest answer to that arithmetic is ground that fights for itself, so the people we send face less of it. That is what this system is for, and it is why it is free.

The specification, hardware annex, and governance charter are published under permissive open license. Build one. Improve it. Propose changes. It belongs to everyone it protects.

The Open Source Firefighter specification family: main specification, Hardware Annex A, governance charter, design philosophy. Published under permissive open license.